

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Worked Solutions

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.**Solution**

1. Use $a^0 = 1$ and $a^{-n} = 1/a^n$; convert $0.2 = 1/5$.

Final answer

$$1, \frac{1}{8}, 25$$

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.**Solution**

1. Add exponents when multiplying, subtract when dividing, and multiply exponents for a power of a power.

Final answer

$$a^3, a^8, \frac{b^2}{a^2}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q003 **Written****Find:** (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.**Solution**1. Apply the zero and reciprocal-power laws; $0.25 = 1/4$.**Final answer** $1, \frac{1}{9}, 16$

Worked Solutions

Laws of Exponents and Roots

Q004 **Written**

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, and $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Solution

1. Collect the exponent of each common base; distribute a negative outer power over every factor.

Final answer

$$a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q005 Written

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.**Solution**

1. Convert terminating decimals to fractions before applying the reciprocal rule.

Final answer $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Worked Solutions

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Solution

1. Use one exponent line for each common base and leave negative powers in reciprocal form when useful.

Final answer

$$a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4, 1$$

Worked Solutions

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, and $(a/2)^3a^{-4} \div (2a)^{-2}$.

Solution

1. Distribute powers first, then combine exponents base by base; keep coefficients separate from variables.

Final answer

$$25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$$

Worked Solutions

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Solution

1. Rewrite the first factor as $(2a^2b^3)^{-2}$; distribute powers and combine all a- and b-exponents.

Final answer

$$\frac{1}{a^7b^6}$$

Q009 Written

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

Solution

1. Recognise $64 = 4^3$ and $1/27 = (1/3)^3$.

Final answer

$$4, \frac{1}{3}$$

Worked Solutions

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[4]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Combine equal-index roots, factor perfect powers, then collect like cube-root terms.

Final answer $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 Written

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Solution

1. Use perfect fourth powers.

Final answer $3, \frac{1}{2}$

Worked Solutions

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Solution

- Combine roots with the same index and extract perfect powers; leave the nested radical exact.

Final answer

4, 2, 10, $\sqrt[3]{4}$, $4\sqrt[3]{3}$

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Solution

- Prime-factorise each radicand and match the index to a perfect power.

Final answer

6, 5, 3, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{3}$, 1, 10, 0.1

Worked Solutions

Laws of Exponents and Roots

Q014 Written

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Solution

1. Convert every radicand to prime factors, combine equal-index roots, and collect only like radicals.

Final answer

$2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Worked Solutions

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Solution

1. Since $4096 = 2^{12}$, both numerator and denominator equal 2.

Final answer

1

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Solution

1. Use $\sqrt[n]{a^m} = a^{m/n}$ and move a radical denominator to a negative exponent.

Final answer $5^{1/2}$, $2^{4/5}$, $3^{-3/5}$

Worked Solutions

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

Solution

1. Rewrite each base as a perfect power, multiply fractional exponents, and apply negative powers last.

Final answer

$$9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$$

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

Solution

1. Index over radicand exponent gives the rational power; reciprocals change the sign.

Final answer

$$a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

Solution

1. Prime-factorise the bases before multiplying the fractional exponents.

Final answer

$$27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$$

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.

Solution

1. Translate each root and reciprocal directly into a rational exponent.

Final answer

$$2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/2)^{-1/3}]^{-2/3}$, $[(1/2)^{-1/3}]^{-2/3}$, $[(1/2)^{-1/3}]^{-2/3}$

Solution

1. Express every base as a power of a small integer, then multiply exponents exactly.

Final answer

$125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Worked Solutions

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Solution

1. Since $1000/27 = (10/3)^3$, the inner power is $(10/3)^{-2}$; multiply the remaining exponents.

Final answer

$$\frac{10}{3}$$

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[5]{32} \times \sqrt[3]{16}$.

Solution

1. Make the bases the same, then add/subtract rational exponents.

Final answer

1, 6, 6, 2

Worked Solutions

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Solution

1. Rewrite powers of 4 and 27 with bases 2 and 3, then combine exponents.

Final answer

5, 8, 7, 9

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Solution

1. Translate every radical to exponents of a and b, then collect like bases.

Final answer

$a^{1/3}$, $b\sqrt{a}$

Worked Solutions

Laws of Exponents and Roots

Q026 **Written**

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Solution

1. Make all bases equal before adding and subtracting rational exponents; extract perfect powers.

Final answer

$$1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$$

Q027 **Written**

For $a > 0, b > 0$, simplify the six given expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[6]{a}$ and mixed a,b radicals.

Solution

1. Convert all radicals to rational powers and combine exponents separately for a and b.

Final answer

$$a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$$

Worked Solutions

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Solution

1. The printed decimal is $0.004 = 1/250$, so the exact result is not rational despite the prompt's wording; retain the exact radical form.

Final answer

$$\left(\frac{2025\sqrt{10}}{64} \right)^{3/2}$$

Q029 Written

Assess the statement $\sqrt{a^2} = a$ **for all real a, using positive and negative examples.**

Solution

1. For $a = 3$ the value is 3; for $a = -3$ the value is 3, not -3.

Final answer

Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Worked Solutions

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Solution**

1. Take the reciprocal of 2^3 .

Correct option: B**Final answer** $1/8$

Q031 MCQ

When multiplying equal bases, we:**Solution**

1. Use $a^m a^n = a^{m+n}$.

Correct option: B**Final answer**

add exponents

Worked Solutions

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:**Solution**1. Subtract: $5 - (-3) = 8$.

Correct option: B

Final answer a^8

Q033 MCQ

A negative exponent means:**Solution**1. Use $a^{-n} = 1/a^n$.

Correct option: B

Final answer

take the reciprocal power

Worked Solutions

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Solution**1. $0.04 = 1/25$, so the reciprocal square is 25^2 .

Correct option: C

Final answer

625

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Solution**1. Compute $-6 + 4 = -2$.

Correct option: A

Final answer

-2

Worked Solutions

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Solution**1. Combine the final powers as $a^{-7}b^{-6}$.

Correct option: B

Final answer

$$1/(a^7b^6)$$

Q037 MCQ

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:**Solution**1. The exponent is $5 - 6 + 1 = 0$.

Correct option: B

Final answer

$$1$$

Worked Solutions

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:**Solution**1. Since $4^3 = 64$.

Correct option: B

Final answer

4

Q039 MCQ

 $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Solution**1. Combine to $\sqrt[3]{216} = 6$.

Correct option: C

Final answer

6

Worked Solutions

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Solution**

1. $(1/2)^4 = 1/16$.

Correct option: A**Final answer**

$$1/2$$

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:**Solution**

1. Collect $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Correct option: B**Final answer**

$$2\sqrt[3]{2}$$

Worked Solutions

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Solution

1. $5^4 = 625$.

Correct option: B

Final answer

5

Q043 MCQ

When combining roots with the same index, multiply:

Solution

1. Use $\sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$.

Correct option: B

Final answer

the radicands

Worked Solutions

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Solution**

1. Both nested roots are 2.

Correct option: B**Final answer**

1

Q045 MCQ

 $\sqrt[5]{2^4}$ in rational-index form is:**Solution**

1. Index becomes the denominator.

Correct option: A**Final answer** $2^{4/5}$

Worked Solutions

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Solution**

1. $(3^3)^{2/3} = 3^2$.

Correct option: C**Final answer**

9

Q047 MCQ

A reciprocal square root is written as:**Solution**

1. A denominator changes the exponent sign.

Correct option: B**Final answer**

$$a^{-1/2}$$

Worked Solutions

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Solution**1. $81^{3/4} = 27$, then take the reciprocal.

Correct option: C

Final answer

1/27

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:**Solution**

1. Radical index is the denominator.

Correct option: A

Final answer

3/4

Worked Solutions

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:**Solution**

1. Use $64/27 = (4/3)^3$ and multiply exponents.

Correct option: C

Final answer

16/9

Q051 MCQ

The 6-3 challenge simplifies to:**Solution**

1. Collect the three rational exponents.

Correct option: B

Final answer

10/3

Worked Solutions

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3}2^{1/2} \div 2^{7/6}$ equals:**Solution**1. The exponent is $4/6 + 3/6 - 7/6 = 0$.

Correct option: B

Final answer

1

Q053 MCQ

 $4^{5/6}2^{4/3}$ equals:**Solution**1. Rewrite $4 = 2^2$, giving exponent 3.

Correct option: C

Final answer

8

Worked Solutions

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:**Solution**1. Compute $3/2 - 1/3 - 5/6 = 1/3$.

Correct option: A

Final answer

$$a^{1/3}$$

Q055 MCQ

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:**Solution**

1. The exponents add to 1.

Correct option: C

Final answer

$$6$$

Worked Solutions

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Solution

1. Translate roots before combining.

Correct option: B

Final answer

combining rational exponents

Q057 MCQ

The printed decimal in the 6-4 challenge is:

Solution

1. Retain the ministry's printed value exactly.

Correct option: B

Final answer

0.004

Worked Solutions

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:**Solution**

1. The principal square root is nonnegative.

Correct option: C**Final answer** $|a|$

Quiz MCQ 1 Concept Quiz · MCQ

Which law gives $a^m a^n = a^{m+n}$?**Solution**

1. Add exponents for multiplication.

Correct option: A**Final answer**

same-base multiplication

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Solution

1. Combine to $\sqrt[3]{125}$.

Correct option: B

Final answer

5

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The value of $64^{-3/2}$ is:

Solution

1. $64^{3/2} = 512$, then take the reciprocal.

Correct option: C

Final answer

1/512

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Solution

1. The principal root is $|a| = -a$ when a is negative.

Correct option: B

Final answer

$$-a$$

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $a^{-3}a^7 \div a^2$.

Solution

1. The exponent is $-3 + 7 - 2 = 2$.

Final answer

$$a^2$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Extract $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Final answer

$$2\sqrt[3]{2}$$

Quiz WRITTEN 7 Concept Quiz · Written

Evaluate $125^{-2/3}$.

Solution

1. $125^{2/3} = 25$, then take the reciprocal.

Final answer

$$1/25$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Solution

1. Use one positive and one negative counterexample.

Final answer

$\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}, \frac{10}{3}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 1, 9, 5, 9, $\sqrt{5}$, 4, 5, 3, 2, 1, 8, $\sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$